

Akka vs. Azure AI Foundry

A comparison for teams building agentic AI

June 2026: Microsoft has renamed the product Microsoft Foundry; this card covers the Foundry Agent Service.

Azure AI Foundry is a collection of Azure services to integrate, not a platform — and its Agent Service has no availability or state-durability SLA. Microsoft's own documentation states the Agent Service has no SLA, no automatic failover, and that the recovery point for agent state "can be total loss." Akka delivers agents, memory, streaming, APIs, and governance as one runtime at a contractual 99.9999% SLA, sub-1-minute RTO, and zero-byte RPO.

No SLA

FOUNDRY AGENT SERVICE
AVAILABILITY

99.9999%

AKKA SLA, WITH
INDEMNITIES

5+

AZURE SERVICES TO STITCH
TOGETHER

Total loss

DOCUMENTED STATE
RECOVERY POINT

DIMENSION	AZURE AI FOUNDRY (MICROSOFT FOUNDRY)	AKKA
What it is	An agent service orchestrating a set of separately provisioned Azure services	A full-stack agentic systems platform on one runtime
Scope	Agent runtime + bring-your-own Cosmos DB, AI Search, Storage, Key Vault, OpenAI, API Management, Monitor — integrated and operated by you	Agents, memory, streaming, APIs, orchestration, observability, and governance on one runtime
Availability SLA	No SLA on Agent Service availability or state durability	99.9999% — entire platform, backed by indemnities
HA / DR	No automatic failover; no built-in DR; no active-active multi-region replication	Active-active HA/DR; sub-1-min RTO; zero-byte RPO
Recovery point	"Can be total loss"; cross-region state lost on failback	Zero-byte RPO; state fully preserved
Governance / EU AI Act	Assessment & mapping across Purview, Content Safety, API Management, Monitor, Entra — not inline enforcement	Aspect-woven runtime enforcement + full pre-production governance
Portability	Azure identity, gateway, and data services; moving off means rebuilding	Any cloud, on-prem, or Akka cloud; portable specs; sovereign cloud
Cost model	Per-token / PTU OpenAI + per-RU Cosmos + per-tier AI Search + gateway + per-GB logging	Shared compute; up to 90% lower infrastructure; fixed annual fee
Certifications	Inherited Azure certifications + Purview Compliance Manager templates	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)

1. A Collection of Services to Integrate, Not a Platform

A production agent on Azure AI Foundry is assembled from multiple, separately provisioned, separately billed Azure services. In the Agent Service Standard deployment mode, Microsoft's own setup documentation requires the customer to bring and operate an Azure Cosmos DB account (minimum 3,000 RU/s), an Azure AI Search resource, an Azure Storage account, and an Azure Key Vault — and a realistic production agent also needs Azure OpenAI, an API Management gateway, and Azure Monitor / Application Insights. ([Standard agent setup](#))

Azure service	Role in an agent	What the customer owns
Azure OpenAI	Model inference	Per-token billing or PTU capacity reservation

Azure Cosmos DB	Agent state / thread memory	Per-RU billing; ≥3,000 RU/s; backup, replication, failover
Azure AI Search	Vector store / retrieval	Per-tier billing; index rebuild on recovery
Azure Storage	Uploaded files, attachments	Redundancy and failover configuration
Azure API Management	Generative AI gateway, load balancing, circuit breaker	Multi-region gateway deployment and policy
Azure Monitor / App Insights	Observability	Per-GB ingestion; multi-region instances

Microsoft's documentation is explicit that the customer operates durability: "Microsoft and you jointly operate the Foundry Agent Service... You own the durability of stateful dependencies (Azure Cosmos DB, Azure AI Search, Azure Storage)." ([HA/DR for Foundry](#)) The integration, the reliability, and the recovery across these seams are the customer's responsibility. Akka delivers agents, durable memory (4ms reads / sub-10ms writes), real-time streaming, an API layer, orchestration, observability, and governance on one runtime — provisioned once, operated as one system, under one SLA.

2. The Agent Service Has No HA/DR Availability SLA

This is stated in Microsoft's own current documentation (HA/DR pages last updated April–May 2026):

- "Agent Service has no availability or state durability Service Level Agreement (SLA)." ([HA/DR for Foundry](#))
- "Foundry itself doesn't provide automatic failover or disaster recovery."
- "Agent Service doesn't provide built-in disaster recovery capabilities. It doesn't replicate state, create backups, or support point-in-time restore... no supported method for active-active, multi-region replication." ([Agent Service DR](#))
- "The recovery point for stateful content can be total loss."

Metric	Azure AI Foundry Agent Service	Akka
Availability SLA	No SLA (availability or state durability)	99.9999%
HA/DR mode	Warm-standby reconstruction; no automatic failover	Active-active, automatic
RTO	"30 or more minutes per project"	Sub-1 minute
RPO	"Can be total loss"	Zero byte
Multi-region	"No supported method for active-active"	Active-active across regions
State on failover	"Standby-region state is permanently lost"	Fully preserved

Microsoft documents that recovery on Azure is reconstruction, not failover: "Warm standby environments start mostly empty. Recovery is reconstruction, not promotion of a hot replica," and recovered agents "have no access to prior threads," with "no state... transferable between regions." ([Platform outage recovery](#))

Azure platform outages are a documented reality

A regional event is not a tail risk for an agent with no availability SLA and a "total loss" recovery point. **Oct 29, 2025** — an Azure Front Door global configuration fault broke routing and authentication across Azure-fronted services for hours. **May 29, 2026** — Azure OpenAI Service had a multi-region outage (failures, timeouts, 5XX) with pronounced impact across Europe and Australia East, from retry amplification cascading across regions. ([Azure status history](#))

3. Cost: Many Meters vs. One Fee

AI systems built on Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required to run the same agentic transaction volume, not list price. On Azure AI Foundry, cost is the sum of independent meters: per-token or PTU-reserved OpenAI, per-RU Cosmos DB (≥3,000 RU/s minimum), per-tier AI Search, gateway units, and per-GB log ingestion.

PTU reservations carry a substantial monthly floor; independent 2026 pricing analyses put a single minimum PTU reservation in the low thousands of dollars per month before the surrounding services are counted ([CloudZero](#)). Each meter scales independently, making the total bill hard to forecast.

Akka runs orchestration, agents, memory, streaming, APIs, observability, and governance on **shared compute** under a **fixed annual fee**. The efficiency is structural: actor-based concurrency delivers ~10 trillion tokens per core per year versus ~2 trillion for comparable solutions, and ~80% less compute than Python-based frameworks. Manulife reported up to **300% more concurrency** and **30–50% faster processing** after porting Python-based systems to Akka. ([akka.io/blog/go-slow-to-go-fast](#))

4. Governance and the EU AI Act: Assessment vs. Inline Enforcement

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9–15)	EUR 15M or 3% global turnover
Supplying incorrect information	EUR 7.5M or 1.5% global turnover

Prohibited-practice penalties enforceable since Feb 2025, high-risk since Aug 2025; high-risk AI carries a 10-year logging-retention obligation (Art. 72).

Microsoft has real assessment tooling: Purview Compliance Manager ingests the EU AI Act, extracts controls with AI, maps improvement actions, and syncs Foundry evaluation results. ([Purview + Foundry](#)) That assesses and documents. The line is architectural: it observes, scores, and maps across several services — it is not a single enforcement plane woven into the agent's execution.

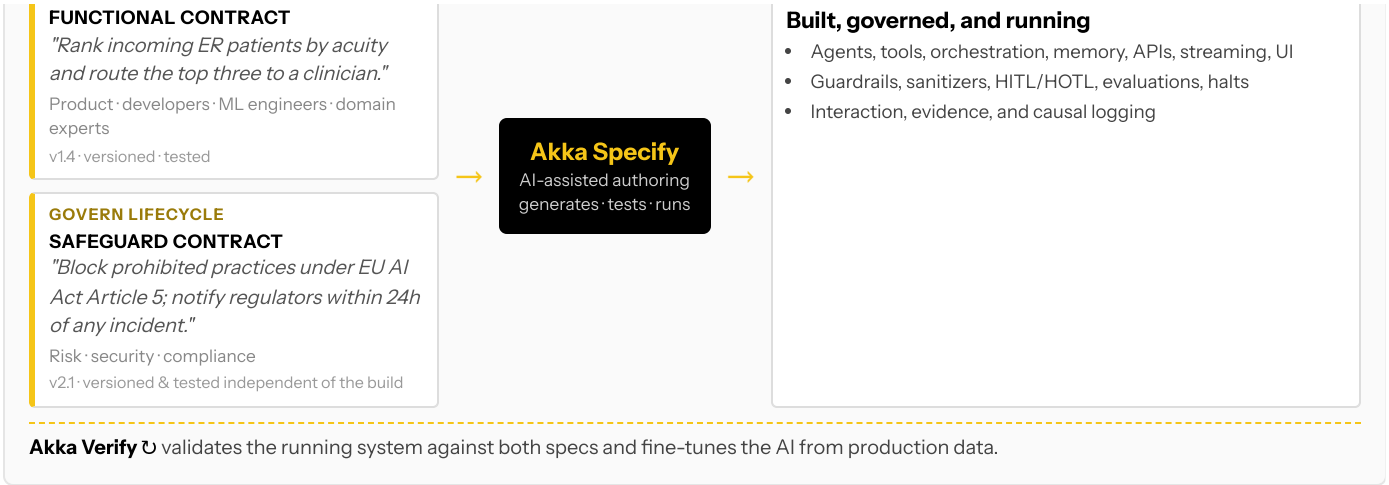
Requirement	Azure approach — the gap
Real-time policy enforcement	Content Safety + API Management; multiple points, not a unified inline engine
Decision explainability	Post-hoc evaluations, not an inline record of a running decision
Immutable interaction log	Standard Monitor/Purview logs, not a hash-chained ledger
Human pause / override	Not built in; a log reader cannot stop a running process
Authorization capture	Correlate Entra + RBAC + gateway logs, not captured atomically
PII scrub with explain	Detection exists; no single atomic scrub-and-explain
Pre-deployment classification + sign-off	Compliance Manager checklists, not a runtime-gating attestation engine
Sealed audit artifact	Assembled from reports; no per-deployment sealed package

Akka's governance is **aspect-woven into the runtime**: inline guardrails, policies, LLMs-as-a-judge, and sanitizers execute within the agents; evidence is hash-chained and immutable; humans can pause, override, or nudge a running process; PII is scrubbed and explained atomically. Before a system ships, Akka classifies it against **186 regulations and 877 controls (288 carrying financial penalties)**, routes change events to the right reviewers, and emits a sealed Governance Posture Package. Akka Verify proves conformance from the running system.

5. Two Lifecycles, One Certified System

Building on Azure AI Foundry means developers assembling agents against Azure SDKs and portals; there is no first-class lifecycle for a risk officer to author and independently version the safeguard contract. Akka runs two independent lifecycles on one platform via **Akka Specify**:





The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences. Azure AI Foundry has no equivalent governance authoring lifecycle that produces a certified, sealed system.

6. Real-Time Streaming at Petabyte Scale

Azure AI Foundry has no built-in streaming engine; real-time pipelines are provisioned separately and operated by the customer. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

7. For the Buyer: A Recent, Renamed Product vs. an 18-Year Platform

Microsoft is a durable company; the question is the maturity of the **product**. Foundry Agent Service reached general availability at Build in May 2025, and the product is mid-rebrand from "Azure AI Foundry" to "Microsoft Foundry," with RBAC roles renamed from "Azure AI User/Owner/Project Manager" to "Foundry User/Owner/Project Manager" as the change rolls out. ([HA/DR for Foundry](#)) It is a young, evolving agent layer assembled over a set of Azure services — and that layer ships today with no availability or state-durability SLA.

Buyer concern	Azure AI Foundry	Akka
Product maturity	GA May 2025; active rebrand (Azure AI Foundry → Microsoft Foundry) and RBAC rename in progress	18 years; 100,000+ production deployments; 52 banks; 2B+ people reached daily
Scope of accountability	Customer integrates and operates Cosmos DB, AI Search, Storage, gateway, OpenAI; Agent Service has no SLA	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard Azure terms; no Agent Service availability/durability SLA	Availability and data-integrity guarantees backed by contractual indemnities
HA / DR	No automatic failover; "total loss" recovery point	Active-active; sub-1-min RTO; zero-byte RPO
Certifications & audits	Inherited Azure certifications + Purview Compliance Manager templates	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Budget predictability	Several independent meters that scale with load	Fixed annual fee on shared compute

The decision is scope and accountability: Azure AI Foundry gives you an agent service to wire your Azure estate around; Akka gives you the integrated platform with the SLA on the whole thing.

Customers Running Agentic and Real-Time Systems on Akka

Manulife Agentic AI selected Akka (Mar 2026) to operationalize agentic AI	Tubi 5B tok/s real-time hyper-personalization engine	Swiggy 71ms order-assignment AI, ~50% latency reduction	John Deere 1,000+ tractor sensors turned into real-time insight	Verizon 750% order-processing capacity gain; 6s → 2.4s response
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Common Questions

We're already on Azure. Why add Akka?

Akka deploys inside your Azure VPC. You keep your Azure infrastructure, your Entra ID, and your networking — Akka runs alongside them and adds the integrated runtime, active-active HA/DR, the 99.9999% SLA, and inline governance the Foundry Agent Service does not provide on its own.

Won't Microsoft close the HA/DR gap soon?

Microsoft's current documentation (updated Apr–May 2026) states the Agent Service has no availability or state-durability SLA, no automatic failover, no built-in DR, and that the recovery point "can be total loss." Until that changes in the product, the availability of an assembled Foundry agent is the customer's responsibility. Akka delivers active-active HA/DR with a contractual SLA today.

Doesn't Microsoft already cover the EU AI Act?

Microsoft has real assessment tooling: Purview Compliance Manager extracts EU AI Act controls, maps improvement actions, and syncs Foundry evaluation results. That assesses and documents; it does not enforce inline. The Act expects immutable records witnessed as they happen, human override of a running process, authorization capture at execution time, and pre-deployment classification that gates a release. Akka enforces these in the runtime and produces a sealed posture package.

Isn't Foundry cheaper because we have an EA?

EA discounts apply to individual Azure services, but a production agent still meters across Azure OpenAI, Cosmos DB, AI Search, Storage, the gateway, and logging — each scaling independently. Akka's shared-compute model runs all of it on one runtime at a fixed annual fee, and is up to 90% cheaper to operate for the same agentic transaction volume.

Sources

Foundry HA/DR (no SLA, no failover): learn.microsoft.com/en-us/azure/foundry/how-to/high-availability-resiliency (updated 2026-04-15) — "Agent Service has no availability or state durability SLA"; "Foundry itself doesn't provide automatic failover or disaster recovery"; customer owns durability of Cosmos DB / AI Search / Storage

Foundry Agent Service DR: learn.microsoft.com/en-us/azure/foundry/how-to/agent-service-disaster-recovery (updated 2026-05-12) — "doesn't replicate state, create backups, or support point-in-time restore"; "no supported method for active-active, multi-region replication"; "recovery point for stateful content can be total loss"

Foundry platform outage recovery: learn.microsoft.com/en-us/azure/foundry/how-to/agent-service-platform-disaster-recovery — RTO "30 or more minutes per project"; "reconstruction, not promotion of a hot replica"; "standby-region state is permanently lost"

Standard agent setup (required services): learn.microsoft.com/en-us/azure/foundry/agents/concepts/standard-agent-setup — Cosmos DB (≥3,000 RU/s), AI Search, Storage, Key Vault required; plus OpenAI, API Management, Monitor

Foundry Agent Service GA: techcommunity.microsoft.com — GA announced at Build, May 2025

Purview + Foundry governance: learn.microsoft.com/en-us/purview/ai-azure-foundry — Compliance Manager extracts EU AI Act controls, maps improvement actions, syncs Foundry evaluations (assessment/mapping)

Azure OpenAI / PTU pricing (third-party): cloudzero.com/blog/azure-openai-pricing; opslyft.com/blog/azure-openai-pricing — PTU minimum reservations and multi-meter cost

Azure outages: azure.status.microsoft.com/en-us/status/history — Oct 29 2025 Front Door global fault; May 29 2026 Azure OpenAI multi-region outage (Europe / Australia East)

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1-min RTO, zero-byte RPO (contractual indemnities); 186 regulations / 877 controls / 288 with penalties; 100,000+ deployments / 18 years; profitable; Dell Technologies Capital

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Manulife: akka.io/blog/manulife-selects-akka-to-operationalize-agentic-ai (March 10, 2026)